LEVEL UP ACADEMY

Class 9 | CBSE / GSEB | Science

Practice Pack: Force and Laws of Motion

Chapter / Topic: Topic-based: use with whichever Class 9 book your students follow

Name: _____ Roll No.: _____ Date: _____

Marks obtained: _____

General instructions:

1. Level 1 checks the ideas, Level 2 is board-style numericals, and Level 3 is Foundation-level challenge.
2. Key relations: p = mv · F = ma = (change in momentum)/time · momentum before = momentum after (when no external force acts).
3. Convert g → kg and km/h → m/s before you substitute.

LEVEL 1: CONCEPTS

Q1. State Newton's first law of motion.

Q2. Why do passengers lurch forward when a moving bus stops suddenly?

Q3. Define momentum and give its SI unit.

Q4. A rubber ball and a stone are the same size. Which has more inertia, and why?

Q5. State Newton's third law of motion with one example.

LEVEL 2: NUMERICALS

Q6. A force of 5 N gives mass m_1 an acceleration of 10 m/s^2 and mass m_2 an acceleration of 20 m/s^2. What acceleration would it give if both masses were tied together?

Q7. A car of mass 1200 kg moving at 90 km/h is slowed down to 18 km/h in 4 s. Find the acceleration, the change in momentum, and the force applied.

Q8. An object of mass 100 kg is accelerated uniformly from 5 m/s to 8 m/s in 6 s. Find its initial and final momentum and the force applied.

Q9. A bullet of mass 20 g is fired from a pistol of mass 2 kg with a velocity of 150 m/s. Find the recoil velocity of the pistol.

Q10. A hockey ball of mass 200 g moving at 10 m/s is hit so that it goes back along its path at 5 m/s. Find the change in its momentum.

LEVEL 3: CHALLENGE (FOUNDATION)

Q11. Two objects of masses 100 g and 200 g move along the same line and direction at 2 m/s and 1 m/s respectively. They collide, and after the collision the first object moves at 1.67 m/s. Find the velocity of the second object.

Q12. Why does a cricketer pull the hands backwards while catching a fast ball? Use the idea of force and time.

Q13. A truck starts from rest and rolls down a slope with constant acceleration, covering 400 m in 20 s. Find its acceleration and the force acting on it if its mass is 7 tonnes.

~ Momentum questions: always choose a positive direction first! ~